MIL-C-60982B(AR)
17 November 1980
SUPERSEDING
MIL-C-60982A(PA)
29 March 1976

MILITARY SPECIFICATION
CASE CARTRIDGE, 40MM, M169
PART FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of case cartridge designated as Case, Cartridge, 40MM, M169.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-45	-	Tape, gummed, paper, reinforced and plain for sealing and servicing
PPP-B-636	-	Boxes, Fiber

MILITARY

MIL-A-8625	-	Anodic coatings for aluminum for aluminum alloys
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

★ U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/479 FSC: 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	-	Packaging, packing and marking for inert ammunition component
MIL-STD-1235	-	Single and Multilevel continuous sampling tables for inspection by attributes

* DRAWINGS (See 6.9)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKING DRAWINGS

8886329	-	Case, cartridge, 40MM, M169

INSPECTION EQUIPMENT DRAWINGS

9202576	-	Tensile strength
9202967	-	Coating thickness measuring instrument
8827872	-	Snap gage

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117-73	-	Standard Method of Salt Spray (Fog) Testing
ASTM E18-67	-	Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E 8-69	-	Tension Testing of Metallic Materials, Methods of

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Part. The part shall comply with all requirements specified on Drawing (Dwg.) 8886329, all associated drawings and with all requirements specified in applicable specifications.

3.3 Hardness. The hardness of the area near the mouth and on the base of the cartridge case shall be 67 ± 7 on the Rockwell 30 T Scale.

3.4 Coating thickness. The thickness of the anodic coating shall be a minimum of .3 mil (.0003 inches).

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirments for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:07Z
Check the source to verify that this is the current version before use.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Case, Cartridge	8886329	25 (5 for salt spray)
Tensile specimen	8886329	6

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

TABLE 1. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60982B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Case Cartridge				SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case Catridge (Dwg. 8886329)				
	Examination for defects	25	-	3.2	4.4.2.1
	Hardness	25	-	3.3	4.5.2
	Salt spray	5	-	3.2	4.5.3
	Tensile properties	6	-	3.2	4.5.5
	Diameter of vent holes	25 *	-	3.2	-

NOTES:

* The 25 cartridge cases chall be sectioned and the diameter of each vent hole be measured.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-60982B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 8886329
4.4.2.1	Case Cartridge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Three (3) or more vent holes blocked or missing		100%	3.2	Automatic Gage or Visual/Manual
Major					
101.	One (1) or two (2) vent holes blocked or missing		100%	3.2	Automatic Gage or Visual/Manual
102.	Diameter of vent holes		0.40%	3.2	Gage
103	Hardness		0.40%	3.3	4.5.2
104.	Burr in vent hole		0.40%	3.2	Visual/Equipment
105.	Salt spray	(a)	(a)	3.2	4.5.3
106.	Tensile properties	6	0-1	3.2	4.5.5
107.	Concentricity of flange diameter with second largest outside diameter	See 4.5.1			
108.	Concentricity of largest inside diameter open end with second largest outside diameter		0.40%	3.2	Gage
109.	Concentricity of base plug cavity with second largest outside diameter		0.40%	3.2	Gage
110.	Concentricity of propellant charge cavity with base plug cavity		0.40%	3.2	Gage
111.	Diameter of propellant charge cavity, straight portion		0.40%	3.2	Gage
112.	Diameter of base plug cavity		0.40%	3.2	Gage
113.	Depth of base plug cavity		0.40%	3.2	Gage
114.	Second largest outside diameter		0.40%	3.2	Gage (see 6.6)
115.	Perpendicularity of open end to second largest outside diameter		0.40%	3.2	Gage (see 6.6)

NOTES:

(a) Sampling and rejection shall be in accordance with MIL-A-8625.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 8886329
4.4.2.1	Case Cartridge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116.	Perpendicularity of base to second largest outside diameter		0.40%	3.2	Gage
117.	Diameter of flange		0.40%	3.2	Gage
118.	Diameter below flange		0.40%	3.2	Gage
119.	Total length		0.40%	3.2	Gage
120.	Largest inside diameter, open end		0.40%	3.2	Gage (see 6.7)
121.	Location of large datum diameter, propellant cavity		0.40%	3.2	Gage
122.	Location of small datum diameter, propellant cavity		0.40%	3.2	Gage
123.	Thickness through node		0.40%	3.2	Gage
124.	Thickness of flange		0.40%	3.2	8827872
125.	Depth of sealing tip		0.40%	3.2	Gage
126.	Location of large datum diameter on node		0.40%	3.2	Gage
127.	Finish of vent holes improper		0.40%	3.2	Gage
128.	Location of vent holes		0.40%	3.2	Gage
129.	Metal defective (see 6.8)		0.40%	3.3	Visual
130.	Process control inspection of both temperatures			3.2	4.5.4
Minor					
201.	Protective coating with bare spot, except anodic rack marks		0.65%	3.2	Visual
202.	Color incorrect		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.

4.4.3.1 Anodic coating thickness. The sample size for batch processes shall be selected in accordance with MIL-STD-105, Inspection Level II, AQL 0.40 percent. The sample size for continuous process shall be selected in accordance with MIL-STD-1235, Inspection Level II, AQL 0.40 percent.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.2 Hardness. The cartridge case shall be subjected to a Rockwell Hardness Test in accordance with ASTM E 18-67. This test is a non-destructive test. Parts so tested may be returned to the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:07Z
Check the source to verify that this is the currentversion before use.

4.5.3 Salt spray. The cartridge case and base plug shall be subjected to the salt spray test in accordance with ASTM B 117-73. This test is a destructive test. Parts so tested shall not be returned to the lot. This test shall be conducted for 240 hours.

4.5.4 Process control inspection of bath temperature for anodic coating. Continuous monitoring inspection shall be conducted for bath temperature requirement specified on the applicable drawings. If the bath temperature for a batch of parts does not comply with the applicable drawing requirement, the batch shall be suitably identified, classed defective and removed from the lot.

4.5.5 Tensile properties of cartridge case. The shape of the specimen for determining tensile strength shall be as specified on equipment Dwg. 9202576. The specimen shall be taken prior to trimming and anodizing from a 3/4 inch wide hoop, 1/4 inch in from the mouth of the cartridge case. The hoop shall be split, unrolled an cut to te required shape. The test shall be conducted in accordance with ASTM E 8. Each specimen must exhibit a minimum tensile strength of 50,000 p.s.i. and a minimum yield strength of 45,000 p.s.i.

4.5.6 Anodic coating thickness. The equipment used to measure the thickness shall be in accordance with equipment Dwg. 9202967 and in accordance with the operating and calibration instructions provided. This test is a non-destructive test. Parts so tested may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (See 6.5.

5.1.1 Level C. The parts shall be preserved and packaged as described below:

5.1.1.1 Case cartridge. Three hundred (300) cases shall be packed in six (6) layers in Type 1, Class 1, 8" flute, 175 pound minimum, domestic class, RSC Style, Specification PPP-B-636, fiberboard box. Inside dimensions of fiber box shall be 17 1/2 x 8 3/4 x 13 5/16 high. Each layer shall be composed of an egg crate type separator in 10 x 5 pattern. Separators shall fit snug in fiber box, length and width. Height of separator to be 1 3/4 inches. A filler 17 x 8 shall be used between layers and on top as required to insure tight packing. A spacer 8 5/8 x 8 5/8 shall be placed between flaps on bottom and on top of fiber box. Material used for separator, fillers and spacers shall be same as that used for fiber box.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:07Z
Check the source to verify that this is the currenversion before use.

5.2 Packing.

5.2.1 Level C. The part shall be packed for shipment as specfiied in 5.1.1.1.

5.3 The construction of the fiber box and the material requirements of the fiber box and the material requirements of the fiber box shall comply with PPP-B-636. When assembled, corrugations of the fiber box shall parallel the fiber box depth. The manufacturer's joint on the fiber box shall be reinforced with paper tape of metal stapled or stitched.

Spacing of staples or stitches shall not average over 1 1/2 inches apart center to center. A manufacturing tolerance of plus or minus 1/8 inch is permissible. The parts manufacturer shall be furnished first article samples of all packing materials so that proper fit of packing material dimensions provide a proper fit of all parts as determined to be necessary by examination of sample packs are permissible.

5.4 Sealing. The fiber box flaps shall be secured with gummed Kraft paper not less than 3 inches wide, complying with Specification PPP-T-45, Type III, Grade B tape. The tape shall be centered the full length of the outer flap seams and extend over each end of the fiber box at least 3 inches. The flap edge seams with the tape extending at least 3 inches along the corners of the fiber box. Tape shall be perperly moistened and sufficiently compressed to insure tight adhesion to the fiber box.

5.5 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use on 40MM cartridges.

6.2 Ordering data. See MIL-A-48078.

* 6.3 Submission of inspection equipment design for approval. See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRDAR-QA-I, ARRADCOM, Dover, NJ 07801.

6.4 Submission of results of contractor conducted examinations and tests. Data shall be submitted in accordance with data item DI-R-1721 on the DD Form 1432 for the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

6.5 Preservation and packaging. Preservation and packaging may be in accordance with MIL-STD-1169 in lieu of that specified herein at the option of the procuring agancy.

6.6 If the requirement cannot be met in the free state, the "go" inside diameter plug gage may be inserted in the cartridge case while inspecting the "go" and "no go" second largest outside diameter. If the "go" inside diameter plug is utilized, the data cards shall be annotated accordingly.

6.7 If the requirment cannot be met in the free state, a confining fixture may be utilized while inspecting the "go" and "no go" largest inside diameter. If the confining fixture is utilized, the data cards shall be annotated accordingly.

6.8 Metal defective. Nick, crack, dent or sharp edge which might interfere with the proper assembly or end item performance of the item.

* 6.9 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

* The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, correction, deletions) from the previous specification were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccurancies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous specification.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1310-A243

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-60982B Case Cartridge, 40MM. M169 Part For

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:07Z
Check the source to verify that this is the current version before use.